

- 6** Determine whether each function represents growth or decay:
- a** $y = 4(1.08)^x$ **b** $y = 7(0.85)^x$
- c** $y = 2e^{-0.3x}$ **d** $y = (\frac{5}{4})^x$

Compound interest

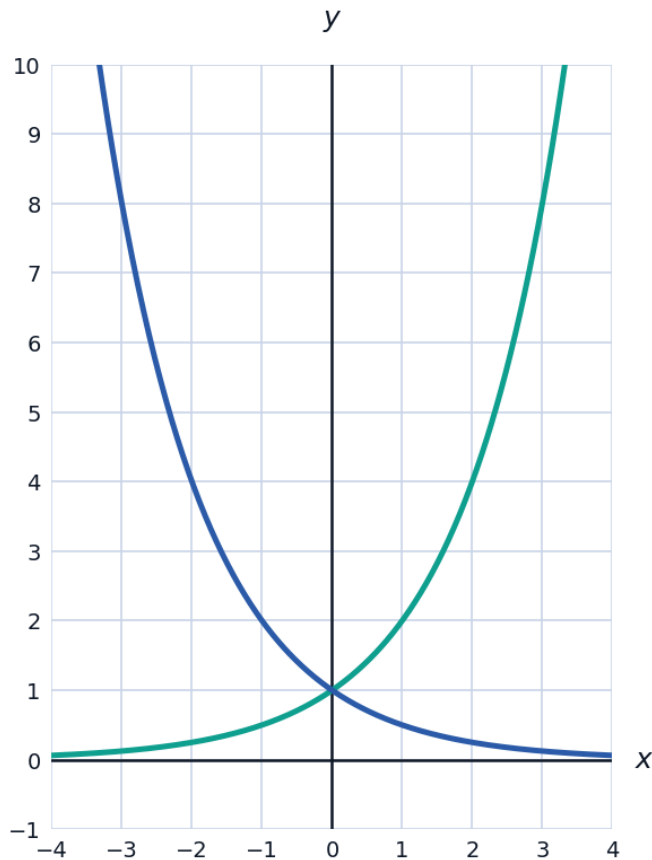
- 7 2000 riyals is invested at 5% annual interest, compounded monthly, using $A = P(1 + \frac{r}{n})^{nt}$.
- a Find the value after 3 years, to the nearest riyal.
- b Find the value after 3 years if compounded continuously instead ($A = Pe^{rt}$), to the nearest riyal.
- 8 How much must be invested now at 4% annual interest compounded annually to have 10,000 riyals in 8 years? Round to the nearest riyal.

Comparing exponential growth rates

- 9 Two investments start at 500 riyals: one grows at 4% per year, the other at 7% per year. Find the value of each after 10 years, to the nearest riyal, and their difference.
- 10 Rank the following functions from slowest to fastest growth as $x \rightarrow \infty$: $y = x^{10}$, $y = 2^x$, $y = 1.01^x$.

Comparing growth and decay graphically

- 11 The graphs of $y = 2^x$ (growth) and $y = \left(\frac{1}{2}\right)^x$ (decay) are shown together. Use the graph to describe the symmetry relating the two curves.



The natural exponential base e

- 12 Explain what distinguishes $e \approx 2.71828$ from other exponential bases, in terms of the slope of $y = e^x$ at $x = 0$.
- 13 Evaluate e^0 , e^1 (to 3 decimal places), and simplify $e^{\ln 5}$.

Piecewise-exponential applications

- 14 A machine depreciates in value according to $V(t) = 15000(0.88)^t$ (riyals, years). Find its value after 5 years, and the year in which its value first drops below 5000 riyals (to the nearest year).

Transformations of exponential functions

15 State the horizontal asymptote and range of each transformed exponential function:

a $y = 3^x + 4$

b $y = -2^x$

c $y = 5^x - 3$

16 Write the equation of the function obtained by taking $y = e^x$, reflecting it across the y -axis, then shifting it down 2 units.

Comparing exponential functions numerically

17 For $f(x) = 3^x$, complete a short table of values at $x = -1, 0, 1, 2, 3$ and verify each value satisfies $f(x + 1) = 3f(x)$.